

## EXPERIMENT – 6

### AIM

To implement modules in Node.js by creating, exporting, importing, and using functions from separate JavaScript files.

### DESCRIPTION

This experiment demonstrates the **module concept in Node.js**. A separate module is created to contain reusable functions, which are exported from the module file and imported into the main JavaScript file. The imported functions are then executed to produce the required output.

### PROGRAM

#### Client.js

```
const os = require('os');

const path = require('path');

const dns = require('dns');

const net = require('net');

const readline = require('readline');

const rl = readline.createInterface({

    input: process.stdin,

    output: process.stdout

});

function osFunction(){

    console.log('platform : '+os.platform());

    console.log('CPU architecture : '+os.arch());

    console.log('CPU information: ',os.cpus());

    console.log('Total memory: '+os.totalmem() +"bytes.");

    console.log('Free memory: '+os.freemem() +"bytes.");

    showMenu();

}
```

```

function pathFunction() {
    rl.question("Enter a file path: ", (filePath) => {
        console.log("Directory: " + path.dirname(filePath));
        console.log("File name: " + path.basename(filePath));
        console.log("Extension: " + path.extname(filePath));
        console.log("Normalized path: " + path.normalize(filePath));
        showMenu();
    });
}

function dnsFunction(){
    rl.question("Enter a domain name: ",(domainname) =>{
        dns.lookup(domainname,(err,address) => {
            if (err){
                console.log("Error: "+err.message);
                return;
            }
            console.log("IP address: "+ address);
            showMenu();
        });
    });
}

function netFunction(){
    const server = net.createServer((socket) => {
        console.log("Client Connected");

        socket.write("Welcome to the Node.js TCP Server!");
        socket.on("data", (data) => {

```

```

        console.log("Message from Client:", data.toString());
    });
    socket.on("end", () => {
        console.log("Client disconnected");
    });
});
server.listen(5000, () => {
    console.log("Server is running on port 5000");
});
}

function showMenu() {
    console.log("\n=====");
    console.log(" Node.js System and Network Information");
    console.log("=====");
    console.log("1. Operating System Information");
    console.log("2. File Path Information");
    console.log("3. DNS Lookup");
    console.log("4. TCP Server");
    console.log("5. Exit");
    rl.question("Enter your choice: ", (choice) => {
        if (choice === "1") {
            osFunction();
        }
        else if (choice === "2") {
            pathFunction();
        }
        else if (choice === "3") {

```

```

        dnsFunction();
    }
    else if (choice === "4") {
        netFunction();
    }
    else if (choice === "5") {
        console.log("Exiting...");
        rl.close();
    }
    else {
        console.log("Invalid choice.");
        showMenu();
    }
});
}
showMenu();

```

### **Server.js**

```

const net = require('net');
const client = net.createConnection(5000, 'localhost', () => {
    console.log('Connected to server');
});
client.on('data', (data) => {
    console.log('Message from server:', data.toString());
    client.end();
});
client.on('close', () => {
    console.log('Connection closed');

```

```
});
```

## DETAILS ABOUT PROGRAM

Folder structure:

FullStack/

```
└── Week-6/
    ├── Client.js
    └── Server.js
```

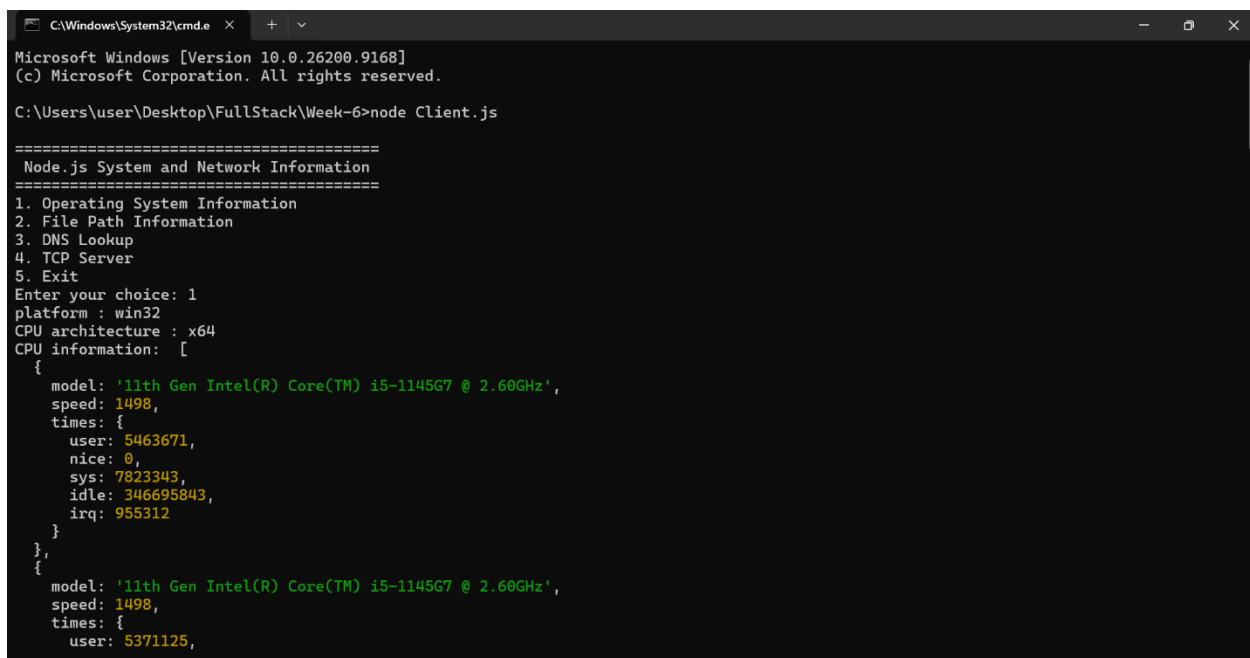
The program is executed using **Node.js in the VS Code Terminal**.

### Execution steps:

1. Open the Node-Modules folder in VS Code.
2. Open the VS Code Terminal.
3. Run the main JavaScript file using Node.js.
4. The module is imported into the main program.
5. The functions from the module are executed.
6. The required output is displayed in the terminal.

## OUTPUT

The Node.js module is successfully created, exported, imported, and executed, and the required output is displayed in the terminal.



```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\user\Desktop\FullStack\Week-6>node Client.js

=====
Node.js System and Network Information
=====
1. Operating System Information
2. File Path Information
3. DNS Lookup
4. TCP Server
5. Exit
Enter your choice: 1
platform : win32
CPU architecture : x64
CPU information: [
  {
    model: '11th Gen Intel(R) Core(TM) i5-1145G7 @ 2.60GHz',
    speed: 1498,
    times: {
      user: 5463671,
      nice: 0,
      sys: 7823343,
      idle: 346695843,
      irq: 955312
    }
  },
  {
    model: '11th Gen Intel(R) Core(TM) i5-1145G7 @ 2.60GHz',
    speed: 1498,
    times: {
      user: 5371125,
```

```
C:\Windows\System32\cmd.e x + v
nice: 0,
sys: 5770015,
idle: 348841671,
irq: 228015
},
{
model: '11th Gen Intel(R) Core(TM) i5-1145G7 @ 2.60GHz',
speed: 1498,
times: {
user: 4155562,
nice: 0,
sys: 3494671,
idle: 352332578,
irq: 119406
},
},
{
model: '11th Gen Intel(R) Core(TM) i5-1145G7 @ 2.60GHz',
speed: 1498,
times: {
user: 3321437,
nice: 0,
sys: 2857890,
idle: 353803484,
irq: 80078
},
},
{
model: '11th Gen Intel(R) Core(TM) i5-1145G7 @ 2.60GHz',
speed: 1498,
times: {
user: 2200734,
```

```
C:\Windows\System32\cmd.e x + v
nice: 0,
sys: 1756500,
idle: 356025578,
irq: 65062
},
{
model: '11th Gen Intel(R) Core(TM) i5-1145G7 @ 2.60GHz',
speed: 1498,
times: {
user: 1716265,
nice: 0,
sys: 1453015,
idle: 356813531,
irq: 43609
},
},
{
model: '11th Gen Intel(R) Core(TM) i5-1145G7 @ 2.60GHz',
speed: 1498,
times: {
user: 1666593,
nice: 0,
sys: 1467437,
idle: 356848781,
irq: 40515
},
},
{
model: '11th Gen Intel(R) Core(TM) i5-1145G7 @ 2.60GHz',
speed: 1498,
times: {
user: 1365765,
```

```
C:\Windows\System32\cmd.e x + v
nice: 0,
sys: 1467437,
idle: 356848781,
irq: 40515
},
{
model: '11th Gen Intel(R) Core(TM) i5-1145G7 @ 2.60GHz',
speed: 1498,
times: {
user: 1365765,
nice: 0,
sys: 1345000,
idle: 357272046,
irq: 29406
},
},
Total memory: 8128958464bytes.
Free memory: 1341857792bytes.
=====
Node.js System and Network Information
=====
1. Operating System Information
2. File Path Information
3. DNS Lookup
4. TCP Server
5. Exit
Enter your choice: 2
Enter a file path: Open the Node-Modules folder in VS Code.
Directory:
File name: Open the Node-Modules folder in VS Code.
```

```
C:\Windows\System32\cmd.e x + v
Extension:
Normalized path: Open the Node-Modules folder in VS Code.
=====
Node.js System and Network Information
=====
1. Operating System Information
2. File Path Information
3. DNS Lookup
4. TCP Server
5. Exit
Enter your choice: Open the VS Code Terminal.
Invalid choice.
=====
Node.js System and Network Information
=====
1. Operating System Information
2. File Path Information
3. DNS Lookup
4. TCP Server
5. Exit
Enter your choice: Run the main JavaScript file using Node.js.
Invalid choice.
=====
Node.js System and Network Information
=====
1. Operating System Information
2. File Path Information
3. DNS Lookup
4. TCP Server
5. Exit
```

```
C:\Windows\System32\cmd.exe
Enter your choice: The module is imported into the main program.
Invalid choice.

=====
Node.js System and Network Information
=====
1. Operating System Information
2. File Path Information
3. DNS Lookup
4. TCP Server
5. Exit
Enter your choice: The functions from the module are executed.
Invalid choice.

=====
Node.js System and Network Information
=====
1. Operating System Information
2. File Path Information
3. DNS Lookup
4. TCP Server
5. Exit
Enter your choice: The required output is displayed in the terminal.
Invalid choice.

=====
Node.js System and Network Information
=====
1. Operating System Information
2. File Path Information
3. DNS Lookup
4. TCP Server
5. Exit
```

```
C:\Windows\System32\cmd.exe
2. File Path Information
3. DNS Lookup
4. TCP Server
5. Exit
Enter your choice: The required output is displayed in the terminal.
Invalid choice.

=====
Node.js System and Network Information
=====
1. Operating System Information
2. File Path Information
3. DNS Lookup
4. TCP Server
5. Exit
Enter your choice: 3
Enter a domain name: www.google.com
IP address: 142.251.150.119

=====
Node.js System and Network Information
=====
1. Operating System Information
2. File Path Information
3. DNS Lookup
4. TCP Server
5. Exit
Enter your choice: 4
Server is running on port 5000
Client Connected
Client disconnected
```

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.26200.9168]
(C) Microsoft Corporation. All rights reserved.

C:\Users\User\Desktop\FullStack\Week-6>node Server.js
Connected to server
Message from server: Welcome to the Node.js TCP Server!
Connection closed

C:\Users\User\Desktop\FullStack\Week-6>
```